

$$1. a. \text{ Number of blocks in main memory} = \frac{\text{Main Memory size}}{\text{Block size}} = \frac{2^{32}}{2^5} = 2^{27} \text{ blocks}$$

$$b. \text{ Number of block in cache} = \frac{\text{Cache size}}{\text{Block size}} = \frac{2^{15}}{2^5} = 2^{10} \text{ blocks}$$

$$c. \text{ Size of offset} = 2^5 = 32 \text{ bytes} \quad] \text{ direct-mapped} \\ = 5 \text{ bits}$$

$$\text{Index field} = 2^{10} = 10 \text{ bits}$$

$$\therefore \text{ tag field} = 32 - (10 + 5) \\ = 32 - 15 \\ = 17 \text{ bits}$$

$$d. \text{ offset field} = 2^5 = 5 \text{ bits} \quad] \text{ fully associative}$$

$$\therefore \text{ tag field} = 32 - 5 \\ = 27 \text{ bits}$$

$$e. \text{ offset field} = 2^5 = 5 \text{ bits}$$

$$\text{number of sets} = \frac{2^{10}}{2^2} = 2^8 = \text{index bits}$$

$$\therefore \text{ tag bits} = 32 - (8 + 5) \\ = 32 - 13 \\ = 19 \text{ bit}$$

$$2. a. \text{ Offset bit} = \log_2 16$$

$$= 4 \text{ bit}$$

$$\text{Index bit} = \log_2 (8)$$

$$= 3 \text{ bit}$$

$$\therefore \text{ tag bits} = 16 - (4 + 3) \\ = 16 - 7 \\ = 9 \text{ bit}$$

b.

Memory address	Tag	Index	Hit / Miss (circle)	
0010 1101 1011 0011	0010 11011	011	Hit	Miss
0000 0110 1111 1100	0000 01101	111	Hit	Miss
0010 1101 1011 1000	0010 11011	011	Hit	Miss
1010 1010 1010 1011	1010 10101	010	Hit	Miss

$$c. \text{ Average access time} = \text{Hit rate of cache} \times \text{Cache access time} + \text{Miss rate of cache} \times \left[\text{Cache access time} + \text{MM access time} \right]$$

$$= (0.95 \times 15) + (0.05 \times (15 + 220))$$

$$= 14.25 + 11.75$$

$$= 26 \text{ ns.}$$

$$3. 1. \text{ offset} = 16 \text{ byte} = 2^4 \text{ byte} = 4$$

$$\begin{aligned} \text{Cache size} &= 4096 \times 2^4 \\ &= 2^{12} \times 2^4 \\ &= 2^{16} \end{aligned}$$

a. Direct-mapped

$$\begin{aligned} \text{tag-bit} &= 64 - 16 \quad \text{index} = 16 - 4 \\ &= 48 \text{ bit} \end{aligned}$$

b. 2-way set associative

$$\begin{aligned} \text{tag bit} &= 64 - (11 + 4) \\ &= 49 \text{ bit} \end{aligned} \quad \begin{aligned} \text{Total sets} &= 4096 / 2 = 2048 \\ &= 2^{11} \end{aligned}$$

c. 4-way set associative

$$\begin{aligned} \text{tag bit} &= 64 - (10 + 4) \\ &= 50 \text{ bit} \end{aligned} \quad \begin{aligned} \text{Total sets} &= 4096 / 4 = 1024 \\ &= 2^{10} \end{aligned}$$

d. fully associative

$$\begin{aligned} \text{tag bit} &= 64 - 4 \\ &= 60 \text{ bit} \end{aligned}$$

$$2. \text{ Ratio} = (\text{tag bit} + \text{valid bit} + \text{data bit}) / \text{data bits}$$

$$a. (48 + 1 + (16 \times 8)) / (16 \times 8) = 177 / 128 = 1.38$$

$$b. (49 + 1 + (16 * 8)) / (16 * 8) = 178 / 128 = 1.39$$

$$c. (50 + 1 + (16 * 8)) / (16 * 8) = 179 / 128 = 1.398$$

$$d. (60 + 1 + (16 * 8)) / (16 * 8) = 189 / 128 = 1.4$$

4. Page size = 64 byte

$$\text{offset} = \log_2(64) = 6 \text{ bits}$$

$$\text{virtual page} = 8$$

$$\text{no. of virtual page} = \log_2(8) = 3 \text{ bits}$$

$$\text{page frame number} = \log_2(4) = 2 \text{ bits}$$

$$a. \text{ bits in virtual address} = 3 + 6 = 9 \text{ bits}$$

$$b. \text{ bits in physical address} = 2 + 6 = 8 \text{ bits}$$

$$c. \text{ bits are virtual page number} = 3 \text{ bits}$$

$$d. \text{ bits are page frame number} = 2 \text{ bits}$$

$$e. \text{ bits are page offset} = 6 \text{ bits}$$

f. 1. 0x00

$$\text{page number} = 000$$

$$\text{offset} = 000000$$

$$\text{Physical address} = 01000000 \\ = 0x40$$

2. 0x44

$$\text{page number} = 001$$

$$\text{offset} = 000100$$

$$\text{Physical address} = 1100 0000 = 0xC0$$

3. 0xC2

$$\text{page number} = 011$$

$$\text{offset} = 000010$$

$$\text{Physical address} = 0000 0010 = 0x02$$

4. 0x80

$$\text{page number} = 010$$

$$\text{offset} = 000000$$

No physical address because of page fault

5.

a. number of pages in virtual memory = $\frac{2^{10}}{2^4} = 2^6$ pages

b. number of pages in physical memory = $\frac{2^8}{2^4} = 2^4$ frames

c. entries in page table:

$$\text{Offset} = \log_2(2^4)$$

$$= 4 \text{ bits}$$

$$\text{total number of bits} = 10 \text{ bits}$$

$$\text{virtual address} = 10 - 4$$

$$= 6 \text{ bits}$$

$$\text{total number of page table entries} = 2^6 \text{ entries}$$

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